MIL-C-24643/28B
3 October 1986
SUPERSEDING
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8 November 1985
MIL-C-24643/28
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1S50MU (INCLUDING VARIATIONS LS1S50MA AND LS1S50MUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be supplied in three variations: LS1S50MU (unarmored), LS1S50MA (armored) and LS1S50MUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, in accordance with ASTM B 286, size 22-7, tin coated. Maximum diameter 0.033 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation, white. Nominal thickness 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Standard identification code applied by method 1.
Fourth - Braided shield of AWG no. 36 tin coated copper. Braid angle of 20 to 25 degrees; minimum coverage of 90 percent.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Nominal diameter over jacketed conductor 0.131 inch.
Sixth - The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from center outward.
Seventh - Binder tape, applied helically with overlap.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LS1S50MU

Eighth - Cross-linked polyolefin jacket. (Nominal thickness 0.050 inch). Cable surface marking required.

Construction variant for type LS1S50MA

Eighth - Cross-linked polyolefin jacket. (Nominal thickness 0.050 inch.)
Ninth - Braided metal armor and paint.

Construction variant for type LS1S50MUS

Eighth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Ninth - Spirally wrapped 0.002-inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
Tenth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Eleventh - Binder tape applied helically with overlap.
Twelfth - Cross-linked polyolefin jacket. (Nominal thickness 0.050 inch). Cable surface marking required.

TABLE I. Details.

Military 1/ part no. M24643/28	Size 2/	Number of conductors	Overall diameter (inches)						Cold bending mandrel diameter maximum (inches)
			Type LS1S50MU		Type LS1S50MA		Type LS1S50MUS		
			min	max	min	max	min	max	
-01UN	16	16	0.760	0.825	0.810	0.875	0.820	0.885	11
-02UN	20	20	.835	.905	.885	.955	.895	.965	12
-03UN	40	40	1.095	1.185	1.145	1.235	1.155	1.245	15
-04UN	70	70	1.465	1.555	1.515	1.605	1.525	1.615	20

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LS1S50MU, LS1S50MA, and LS1S50MUS.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum...	17.54
Voltage withstand - volts, root mean square, minimum	
Conductor to shield.....	5000
Shield to shield (single shields only).....	500
Insulation resistance - shield to shield (LS1S50MUS only, overall shields only), volts, minimum.....	6.0
megohms/1000 feet, minimum.....	1.0

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EXAMINATION AND TESTS: (Continued)

	Requirements
Jacket flaws	No failure
Conductor and shield (LS1S50MUS only) continuity	No failure
Group A:	
Visual and dimensional	No failure
Characteristic impedance - at 1 MHz, ohms	50 ± 2
Group B:	
Attenuation - at 400 MHz, decibel (dB)/100 feet, maximum	18
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - at 0 ± 2°C (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - (LS1S50MU and LS1S50MUS only) - cycles, minimum	250
Shield conformance to material, construction and coverage	No failure
Armor (LS1S50MA only) conformance to material, construction and coverage	No failure
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
(B) Suface transfer impedance (type LS1S50MUS only) Milliohms per meter, maximum	700
EMP response time - ms/500 feet	60

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

		Requirements
	Aging and compatibility (cable) (125 ± 5°C).....	No failure
(B)	Abrasion resistance (jacket) - scrapes, minimum.....	75
	Acid gas equivalent - percent, maximum	
	Jacket.....	2
	Fillers.....	2
	Insulation.....	18
	Halogen content - percent, maximum	
	Jacket.....	0.2
	Fillers.....	0.2
	Flame propagation (cable).....	No failure
	Immersion (jacket)	
	Tensile strength - percent of unaged, minimum.....	50
	Elongation - percent of unaged, minimum.....	50
	Smoke index, maximum	
	Jacket.....	25
	Fillers.....	—
(B)	Insulation.....	45
	Toxicity index, maximum	
	Jacket.....	5
	Fillers.....	5
	Insulation.....	1.5
	Durometer (jacket) - (type A) hardness, minimum.....	80
	Weathering (jacket).....	No failure

UNIT ORDERING LENGTHS:

All sizes 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-28)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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